31/12/2020

Aakash
Medical | IIT-JEE | Foundations
(Divisions of Aakash Educational Services Limited)
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MM : 720 **NEET BOOSTER TEST SERIES (NBTS) for NEET-2021** **Time : 3.00 Hrs.**

Test - 3

Topics covered :

Physics : System of Particles & Rotational Motion, Gravitation
Chemistry : Thermodynamics, Equilibrium
Botany : Anatomy of Flowering Plants, Cell : The Unit of Life
Zoology : Digestion and Absorption, Breathing and Exchange of Gases

Instructions :

(i) Use blue/black ballpoint pen only to darken the appropriate circle.
(ii) Mark should be dark and should completely fill the circle.
(iii) Dark only one circle for each entry.
(iv) Dark the circle in the space provided only.
(v) Rough work must not be done on the Answer sheet and do not use **white-fluid** or any other **rubbing material** on Answer sheet.
(vi) Each question carries 4 marks. For every wrong response 1 mark shall be deducted from total score.

PHYSICS

Choose the correct answer :

1. Two blocks each of 3 kg mass are at rest. Application of forces $\vec{F}_1 = 12\hat{i}$ N and $\vec{F}_2 = 6\hat{j}$ N acting on both masses individually will give velocity of centre of mass at t = 4 s is equal to

(1) $(4\hat{i} + 4\hat{j})$ m/s (2) $(8\hat{i} + 4\hat{j})$ m/s

(3) $(4\hat{i} + 2\hat{j})$ m/s (4) $(8\hat{i} + 8\hat{j})$ m/s

2. Moment of inertia of a disc is minimum
(1) about its geometric axis
(2) about a diameter
(3) about a tangent in its plane
(4) about tangent perpendicular to its plane

3. The ratio of accelerations, when a hollow sphere first rolls down a rough incline and then on a smooth inclined plane of same inclination, is

(1) $\frac{5}{7}$ (2) $\frac{3}{5}$

(3) $\frac{2}{3}$ (4) $\frac{1}{2}$

4. A solid sphere, a hollow sphere and disc, all having same mass and radius, are placed at the top of an incline and released. The friction coefficient between the objects and the incline are same and not sufficient to allow pure rolling, then
(1) The solid sphere take least time to reach the bottom
(2) The hollow sphere take least time to reach the bottom
(3) The disc take least time to reach the bottom
(4) All will take same time to reach the bottom

5. A heavy disc is thrown on a horizontal rough surface with initial velocity v without rotating. What will be the speed, when it starts pure rolling motion?

(1) $\frac{2v}{3}$ (2) $\frac{3v}{5}$

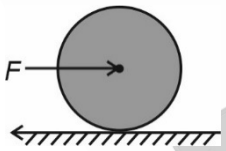
(3) $\frac{v}{3}$ (4) $\frac{v}{2}$

6. A solid cylinder of mass 'M' rolls down without slipping on an inclined plane making an angle 45° with the horizontal. The minimum value of frictional coefficient between the cylinder and incline plane is

(1) $\frac{1}{2}$ (2) $\frac{1}{3}$

(3) $\frac{3}{4}$ (4) $\frac{1}{4}$

7. A force F is applied at the centre of solid sphere of mass m and radius R placed over a rough horizontal plane. The minimum value of coefficient of friction between sphere and the surface for pure rolling is



(1) $\frac{2F}{7\,mg}$ (2) $\frac{3F}{7\,mg}$

(3) $\frac{2F}{5\,mg}$ (4) $\frac{3F}{5\,mg}$

8. A flywheel of moment of inertia 5×10^{-2} kg m^2 is rotating with an angular velocity of 40 rad/s. The constant torque required to bring it to rest in 5 s is

(1) 0.8 Nm (2) 1 Nm

(3) 0.4 Nm (4) 2 Nm

9. When 400 J of work is done on a flywheel its frequency of rotation increases from 3 Hz to 5 Hz. The moment of inertia of the wheel about the axis of rotation is (Take $\pi^2 = 10$)

(1) 3.2 kg m^2

(2) 0.4 kg m^2

(3) 1.25 kg m^2

(4) 25 kg m^2

10. Two identical discs are placed one over the other and both are rotating about a common axis through their centre and normal to their plane with angular velocity ω. If the upper disc is suddenly removed, the angular speed of the other disc will be

(1) 4ω (2) $2\,\omega$

(3) $\omega/2$ (4) ω

11. A ring is in pure rolling motion. In rolling motion ring possess translational kinetic energy (K_t) as well as rotational kinetic energy (K_r). The value of $K_t : (K_t + K_r)$ for the ring is

(1) 7 : 10 (2) 5 : 7

(3) 2 : 5 (4) 1 : 2

12. Area of a parallelogram formed by vectors $\vec{A} = (3\hat{i} - \hat{j} + \hat{k})$ m and $\vec{B} = (\hat{i} + \hat{j} + 2\hat{k})$ m is

(1) 25 m^2 (2) 15 m^2

(3) $15\sqrt{2}$ m^2 (4) $25\sqrt{2}$ m^2

13. When a mass is rotating in a plane about a fixed point, Its angular velocity is directed along

(1) The radius

(2) The tangent to the orbit

(3) A line perpendicular to the plane of rotation

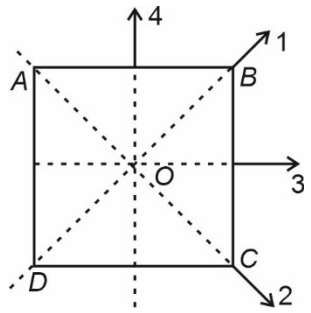
(4) The line making an angle 45° to the plane of rotation

14. Which of the following vector relation is incorrect? (where symbols have their usual meaning)

(1) $\vec{v} = \vec{\omega} \times \vec{r}$ (2) $\vec{\tau} = \vec{F} \times \vec{r}$

(3) $\vec{L} = \vec{r} \times \vec{P}$ (4) $w = \vec{F} \cdot \vec{d}$

15. The moment of inertia of a thin square plate $ABCD$ (shown in the figure) of uniform thickness about an axis passing through centre O and perpendicular to the plate is equal to



(1) $I_1 + I_2$ (2) $I_1 + I_3$

(3) $I_3 + I_4$ (4) All of these

16. Two discs *A* and *B* of same moment of inertia *I* rotating about their regular axis passing through centre and perpendicular to the plane of disc with angular velocity ω and 2ω in same sense respectively. They are brought in contact face to face coinciding the axis of rotation. The expression for loss of kinetic energy during this process is

(1) $\frac{I\omega^2}{2}$ (2) $\frac{2I\omega^2}{3}$

(3) $\frac{I\omega^2}{4}$ (4) $\frac{I\omega^2}{3}$

17. A light rod of length 2ℓ has two masses *m* and $2m$ attached to its two ends. The moment of inertia of the system about an axis making an angle 30° with rod length and passing through the centre of mass of the rod will be

(1) $\frac{m\ell^2}{3}$ (2) $\frac{2m\ell^2}{3}$

(3) $\frac{3m\ell^2}{4}$ (4) $\frac{m\ell^2}{4}$

18. Three masses 1 kg, 3 kg and 4 kg are placed at point *A*(0, 0) m, *B*(0, 2) m and *C*(0, 1) m, the distance of centre of mass of the system from the origin will be

(1) 1.5 m (2) 2.25 m

(3) 1.25 m (4) 0.75 m

19. A thin circular ring of mass $3m$ and radius *R* is rotating in a horizontal plane about an axis vertical to its plane with a constant angular velocity ω. If two objects of mass *m* and $2m$ be attached gently to opposite ends of a diameter of the ring, the ring will then rotate with angular velocity of

(1) $\frac{\omega}{4}$ (2) $\frac{\omega}{2}$

(3) $\frac{3\omega}{4}$ (4) $\frac{\omega}{3}$

20. The ratio of the radii of gyration of a hollow sphere to the solid sphere, both of same mass and radius, about their respective geometric axis is

(1) $\sqrt{5}:\sqrt{3}$ (2) $\sqrt{2}:\sqrt{3}$

(3) $\sqrt{5}:\sqrt{2}$ (4) 1 : 1

21. From a uniform disc of radius *R*, a smaller disc of radius $\frac{R}{2}$ is removed. The centre of the hole formed is at $\frac{R}{2}$ from the centre of the original disc. The shift in position of centre of mass of the disc after removing smaller disc will be

(1) $\frac{R}{3}$ (2) $\frac{R}{2}$

(3) $\frac{R}{4}$ (4) $\frac{R}{6}$

22. A hollow sphere rolls down on two different inclined planes of the same height but different angle of inclination, choose the correct statement.

(1) Acceleration of centre of mass will be same on both inclined plane

(2) On both inclined, hollow sphere take same time to reach the bottom

(3) Speed of the hollow sphere will be same on reaching the bottom of the both inclined planes

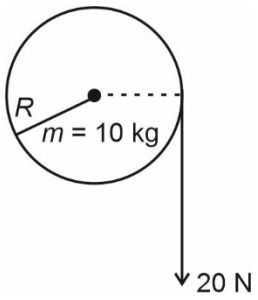
(4) Both (2) and (3)

23. A solid sphere rolls up a rough inclined plane of angle of inclination 30°. At the bottom of the inclined plane, the centre of mass of the solid sphere has a speed of 10 m/s. The maximum displacement of the solid sphere on the inclined will be (g = 10 m/s^2)

(1) 7 m (2) 9 m

(3) 14 m (4) 18 m

24. A cord of negligible mass is wound round the rim of a flywheel (disc) of mass 10 kg and radius 10 cm. A steady pull of 20 N is applied on the cord as shown in figure. The flywheel is mounted on a horizontal axle with frictionless bearings. The angular acceleration of flywheel (disc) is



(1) 30 rad/s^2 (2) 40 rad/s^2

(3) 20 rad/s^2 (4) 25 rad/s^2

25. A thin uniform rod of length L and mass M is suspended from one of its ends. It is set into oscillations about a horizontal axis. Its angular speed is ω while passing through its mean position. How high will its centre of mass rise from the lowest position?

(1) $\frac{\omega^2 L^2}{3g}$ (2) $\frac{\omega^2 L^2}{6g}$

(3) $\frac{\omega^2 L^2}{\sqrt{3}g}$ (4) $\frac{\omega^2 L^2}{2g}$

26. A particle performs uniform circular motion with an angular momentum L. If the frequency of particle motion is doubled and its kinetic energy halved, the angular momentum becomes

(1) 2 L (2) 4 L

(3) $\frac{L}{2}$ (4) $\frac{L}{4}$

27. The position of a particle is given by $\vec{r} = \hat{i} + 2\hat{j} - \hat{k}$ and its momentum is given by $\vec{P} = 3\hat{i} + 4\hat{j} - 2\hat{k}$. The angular momentum of this particle is perpendicular to

(1) x-axis

(2) y-axis

(3) z-axis

(4) Line at equal angles with all the three axis

28. If R is radius of the earth, the amount of work done in lifting a body of mass m from the surface of earth to a height equal to two times the radius of the earth is (g is acceleration due to gravity on surface)

(1) $\frac{mgR}{3}$ (2) $\frac{2mgR}{3}$

(3) $\frac{mgR}{4}$ (4) $\frac{5mgR}{4}$

29. The ratio of escape velocity at earth (V_e) surface to the escape velocity at a planet (V_P) whose radius and mean density are thrice that of earth, is

(1) 1 : 3 (2) 1 : $3\sqrt{3}$

(3) 1 : $\sqrt{3}$ (4) 1 : 1

30. If radius of the earth suddenly shrinks to half of its present value, mass remaining unchanged, then acceleration due to gravity on its surface in comparison to the original, will becomes

(1) The same (2) Half

(3) Double (4) Quadruple

31. The acceleration due to gravity at depth 2 km below the surface of earth is same as at height h above the surface of earth then

(1) h = 1 km (2) h = 2 km

(3) h = 1/2 km (4) h = 3 km

32. If R is radius of the earth, g is mean acceleration due to gravity on its surface, mean density of the earth would be

(1) $\frac{3g}{4\pi RG}$ (2) $\frac{4g}{3\pi RG}$

(3) $\frac{4G}{3\pi Rg}$ (4) $\frac{3G}{4\pi Rg}$

33. Pick out the correct statement regarding acceleration due to gravity

(1) It is more at equator than at poles

(2) It is not affected by the rotation of earth

(3) It is affected by the rotation of the earth

(4) It is not affected by the latitude

34. A satellite is moving very close to a planet of density ρ. The time period of the satellite is

(1) $\sqrt{\frac{3\pi}{\rho G}}$ (2) $\left(\frac{3\pi}{\rho G}\right)^{3/2}$

(3) $\sqrt{\frac{3\pi}{2\rho G}}$ (4) $\left(\frac{3\pi}{2\rho G}\right)^{3/2}$

35. The kinetic energy of a satellite of mass m revolving around the earth at a height equal to radius of earth (R) is

(1) mgR (2) $2mgR$

(3) $\frac{mgR}{2}$ (4) $\frac{mgR}{4}$

36. A man is having mass 75 kg on the surface of the earth. His weight in a geostationary satellite is

(1) Infinity (2) 15 N

(3) 75 N (4) Zero

37. For a satellite moving in an circular orbit around the earth, the ratio of kinetic energy to the magnitude of total energy of the satellite is

(1) 1 : 2 (2) $1:\sqrt{2}$

(3) 1 : 1 (4) 1 : 3

38. A particle of mass m is situated at the centre of a spherical shell of same mass and radius a. The gravitational potential at a point which is at $\frac{3a}{2}$ distance from the centre of the shell, will be

(1) $-\frac{3Gm}{a}$ (2) $-\frac{2Gm}{a}$

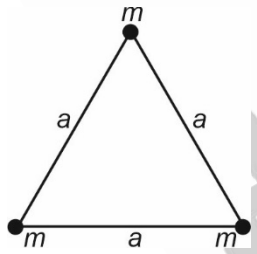
(3) $-\frac{4Gm}{a}$ (4) $-\frac{4Gm}{3a}$

39. A geostationary satellite is orbiting the earth at height of $6R$ above the surface of the earth. The time period of another satellite in hour at height 2.5 R from the surface of earth is

(1) 12 h (2) $6\sqrt{2}$ h

(3) 4 h (4) $4\sqrt{2}$ h

40. Three particles each of mass m are placed at the vertices of triangle of side a as shown in the figure. The gravitational potential at the centroid of the triangle will be



(1) $-\frac{3Gm}{a}$ (2) $-\frac{3\sqrt{3}Gm}{a}$

(3) $-\frac{Gm}{a}$ (4) $-\frac{Gm}{\sqrt{3}a}$

41. A planet orbits the sun in an elliptical orbit. The planet has a constant [Ignore cosmic dust]

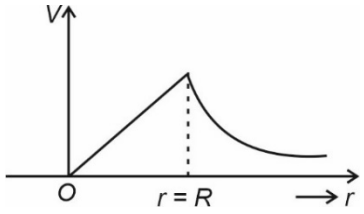
(1) Angular speed

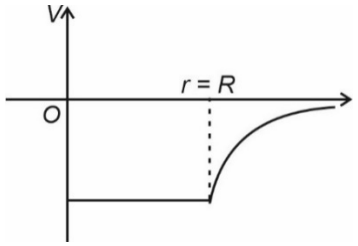
(2) Kinetic energy

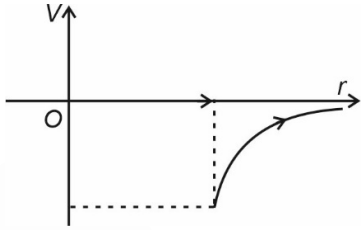
(3) Linear speed

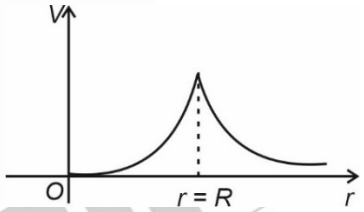
(4) Total mechanical energy

42. Which one of the following plots represents the variation of gravitational potential with distance r due to a thin spherical shell of radius R? (r is measured from the centre of the spherical shell)

(1)


(2)


(3)


(4)


43. The binding energy of a system of earth and its satellite orbiting around the earth in a circular orbit is E. If M is mass of that satellite then its linear speed in that orbit will be

(1) $\frac{2E}{M}$ (2) $\frac{E}{M}$

(3) $\sqrt{\frac{2E}{M}}$ (4) $\sqrt{\frac{E}{M}}$

44. Newton's law of gravitation is universal because

(1) Force is always attractive

(2) Force acts on all heavenly bodies and particles

(3) Force acts on all light bodies and particles

(4) Force acts on the masses kept at a distance and is not affected by the medium

45. With what velocity should a particle be projected from earth so that its maximum height reach becomes equal to half of radius of earth?

(1) $\sqrt{\frac{gR}{3}}$ (2) $\sqrt{2gR}$

(3) $\sqrt{\frac{2gR}{3}}$ (4) $\sqrt{\frac{gR}{2}}$

CHEMISTRY

46. Which among the following is not a state function?
(1) ΔH (2) ΔU
(3) w (4) ΔG

47. 10 J heat is absorbed by the system during the expansion and the work done by the system is 30 J. The ΔU of the system is
(1) 10 J (2) 20 J
(3) – 10 J (4) – 20 J

48. ΔH is equal to heat absorbed by the system for
(1) Isothermal process
(2) Adiabatic process
(3) Isobaric process
(4) Isochoric process

49. 1 mol of an ideal gas is compressed reversibly and isothermally from 10 L to 1 L at 300 K. The work done during the process is
(1) – 5.74 kJ (2) 5.74 kJ
(3) 56.7 J (4) – 56.7 J

50. Intensive property among the following is
(1) Internal energy
(2) Volume
(3) Heat capacity
(4) Molar heat capacity

51. For which of the following reactions, $\Delta H = \Delta U$?
(1) $CH_4(g) + 2O_2(g) \longrightarrow CO_2(g) + 2H_2O(l)$
(2) $N_2(g) + O_2(g) \longrightarrow 2NO(g)$
(3) $N_2O_4(g) \longrightarrow 2NO_2(g)$
(4) $PCl_5(g) \longrightarrow PCl_3(g) + Cl_2(g)$

52. A process is always spontaneous if
(1) $\Delta H < 0$ and $\Delta S > 0$
(2) $\Delta H < 0$ and $\Delta S < 0$
(3) $\Delta H > 0$ and $\Delta S < 0$
(4) $\Delta H > 0$ and $\Delta S > 0$

53. If heat of formation of SO_2 is – 300 kJ mol^{-1} then, the heat evolved by the combustion of 8 g of S is
(1) 75 kJ (2) 150 kJ
(3) 300 kJ (4) 600 kJ

54. If heat of combustion of cyclohexane, cyclohexene and H_2 are – x, – y and – z kJ mol^{-1} respectively then heat of hydrogenation of cyclohexene (in kJ mol^{-1}) is
(1) x + y – z (2) x – y – z
(3) z – x – y (4) y – x – z

55. Enthalpy change for the reaction,
$CO(g) + \frac{1}{2}O_2(g) \longrightarrow CO_2(g)$ is called
(1) Enthalpy of formation
(2) Enthalpy of combustion
(3) Enthalpy of vaporisation
(4) Enthalpy of atomization

56. If enthalpy for the reaction $CCl_4(g) \longrightarrow C(g) + 4Cl(g)$ is 320 kJ mol^{-1} then average bond enthalpy of C – Cl bond is
(1) 64 kJ mol^{-1} (2) 80 kJ mol^{-1}
(3) 128 kJ mol^{-1} (4) 160 kJ mol^{-1}

57. Enthalpy change on heating 1 mol ice from –10°C to water at 10°C is
(Given : $\Delta_{fus}H$ = 6 kJ mol^{-1} at 0°C,
$C_P[H_2O(l)]$ = 75 J mol^{-1} K^{-1},
$C_P[H_2O(s)]$ = 35 J mol^{-1} K^{-1})
(1) 1.1 kJ (2) 4.9 kJ
(3) 7.1 kJ (4) 8.2 kJ

58. ΔH and ΔS for a reaction are 400 kJ mol^{-1} and 2 kJ K^{-1} mol^{-1} respectively. At what temperature the reaction will be at equilibrium?
(1) 100 K (2) 200 K
(3) 300 K (4) 400 K

59. Select the correct relation
(1) $\Delta_rG° = -2.303\ RT \log K$
(2) $\Delta_rG° = 2.303\ RT \log K$
(3) $\Delta_rG° = \frac{2.303}{RT}\log K$
(4) $\Delta_rG° = \frac{-2.303}{RT}\log K$

60. In which of the following reactions, entropy change is positive?

(1) $C(s) + O_2(g) \longrightarrow CO_2(g)$

(2) $BaCl_2(s) + 2H_2O(l) \longrightarrow BaCl_2.2H_2O(s)$

(3) $2P(s) + 3Cl_2(g) \longrightarrow 2PCl_3(l)$

(4) $CaCO_3(s) \longrightarrow CaO(s) + CO_2(g)$

61. When the temperature of any pure crystalline substance approaches zero Kelvin then which property of solid also approaches to zero according to third law of thermodynamics?

(1) Internal energy (2) Gibbs energy

(3) Entropy (4) Enthalpy

62. Born Haber cycle is used to determine

(1) Gibbs energy

(2) Lattice energy

(3) Entropy

(4) Internal energy

63. If $\Delta_r H$ for the reactions $A \rightarrow B$, $B \rightarrow C$, $C \rightarrow D$ and $A \rightarrow D$ are x, y, z, and w respectively then select the correct relation.

(1) x = y + z + w (2) y = x + z + w

(3) w = x + y + z (4) z = x + y + w

64. The amount of heat released by the complete neutralisation of 1 mol of H_2SO_4 with a strong base is

(1) 13.7 kcal

(2) 27.4 kcal

(3) 20.55 kcal

(4) 6.85 kcal

65. Select the incorrect statement among the following for equilibria involving physical processes.

(1) Equilibrium is possible only in a closed system at a given temperature

(2) Both the opposing processes stop at equilibrium

(3) At equilibrium, there is a dynamic but stable condition

(4) During equilibrium, all measurable properties of the system remain constant

66. Equilibrium constant for the reaction,

$2A(g) + 3B(g) \rightleftharpoons C(g) + D(g)$ is given by the expression

(1) $\frac{[A][B]}{[C][D]}$ (2) $\frac{[A]^3[B]^2}{[C][D]}$

(3) $\frac{[C][D]}{[A]^2[B]^3}$ (4) $\frac{[C][D]}{[2A][3B]}$

67. If K_C for the reaction, $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$ is x then K_C for the reaction,

$NH_3(g) \rightleftharpoons \frac{1}{2}N_2(g) + \frac{3}{2}H_2(g)$ is

(1) x (2) $\frac{x}{2}$

(3) $\frac{1}{\sqrt{x}}$ (4) $\sqrt{x}$

68. Ratio of K_P to K_C for the reaction,

$4NH_3(g) + 5O_2(g) \rightleftharpoons 4NO(g) + 6H_2O(g)$

at T K is

(1) $(RT)^1$ (2) $(RT)^{-1}$

(3) $(RT)^2$ (4) $(RT)^{-5}$

69. For the reaction, $A(g) + B(g) \rightleftharpoons C(g)$; if initially $P_A = P_B = 2$ atm and $P_C = 0$ atm then total pressure at the equilibrium is ($K_P = 1$ atm^{-1})

(1) 1 atm (2) 2 atm

(3) 3 atm (4) 4 atm

70. Among the following for which value of K_C, the extent of reaction is maximum?

(1) 1 (2) 10

(3) 100 (4) 1000

71. Which of the following factors will shift the equilibrium, $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$ + Heat; in the forward direction?

(1) Addition of $SO_3(g)$

(2) Increase in pressure

(3) Increase in temperature

(4) Addition of catalyst

72. Unit of equilibrium constant K_P for the reaction $CO_2(g) + C(s) \rightleftharpoons 2CO(g)$ is

(1) $(atm)^0$ (2) $(atm)^{-1}$

(3) (atm) (4) $(atm)^2$

73. Example of homogeneous equilibrium is

(1) $CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$

(2) $H_2O(g) + CO(g) \rightleftharpoons CO_2(g) + H_2(g)$

(3) $C(s) + H_2O(g) \rightleftharpoons CO(g) + H_2(g)$

(4) $Ni(s) + 4CO(g) \rightleftharpoons Ni(CO)_4(g)$

74. Select the correct statement regarding catalyst

(1) It increases the rate of forward reaction but decreases the rate of backward reaction

(2) It increases the rate of backward reaction but decreases the rate of forward reaction

(3) It increases the rate of forward as well as backward reactions

(4) It decreases the rate of forward as well as backward reaction

75. Conjugate base of $H_2PO_4^-$ ion is

(1) H_3PO_4 (2) HPO_4^{2-}

(3) PO_4^{3-} (4) $H_2PO_3^-$

76. A 1M monobasic acid is 0.001% ionized. The value of acid dissociation constant is

(1) 10^{-2} (2) 10^{-5}

(3) 10^{-10} (4) 10^{-12}

77. Salt which cannot undergo salt hydrolysis is

(1) CH_3COONa (2) NH_4Cl

(3) CH_3COONH_4 (4) NaCl

78. Relation between K_{sp} and s for a salt, AB_2 is

(1) $K_{sp} = s^2$ (2) $K_{sp} = s^3$

(3) $K_{sp} = 4s^3$ (4) $K_{sp} = \frac{s^3}{4}$

79. pH of solution containing 0.2 M CH_3COOH and 0.2 M CH_3COONa is (pK_a of CH_3COOH = 4.7)

(1) 4 (2) 4.7

(3) 5.3 (4) 6.9

80. The solubility of AgCl in 0.1 M KCl solution is (K_{sp} of AgCl = 1.2×10^{-10})

(1) 1×10^{-5} M (2) 1.2×10^{-8} M

(3) 1.2×10^{-9} M (4) 4×10^{-5} M

81. Most acidic compound among the following is

(1) CH_4 (2) NH_3

(3) H_2O (4) HF

82. The pH of 0.1 M ammonium acetate solution at 298 K is (pK_a of CH_3COOH = 4.7, pK_b of NH_4OH = 4.7)

(1) 6.9 (2) 7

(3) 7.1 (4) 13.9

83. If K_b of NH_3 is 2×10^{-5} then K_a for NH_4^+ ion is

(1) 2×10^{10} (2) 5×10^{-10}

(3) 2×10^{-19} (4) 5×10^{-19}

84. For a polyprotic acid, select the correct relation

(1) $K_{net} = K_1 + K_2 + \ldots K_N$

(2) $K_{net} = K_1 \times K_2 \times \ldots \times K_N$

(3) $K_{net} = \frac{K_1 \times K_2 \times K_3}{N} \ldots \times K_N$

(4) $K_{net} = \frac{K_1 + K_2 + \ldots K_N}{N}$

85. Standard enthalpy of formation is zero for

(1) C(diamond) (2) S(monoclinic)

(3) $O_3(g)$ (4) C(graphite)

86. If 1 g water takes 2.44 kJ of heat for evaporation then enthalpy of vapourisation of water (in kJ mol^{-1}) is nearly

(1) 24 (2) 44

(3) 22 (4) 46

87. For an isothermal expansion of an ideal gas, which of the following is not true?

(1) $\Delta U = 0$ (2) $\Delta H = 0$

(3) $\Delta T = 0$ (4) $\Delta S = 0$

88. Value of equilibrium constant for a particular reaction depends upon

(1) Concentration of reactants and products only

(2) Temperature only

(3) Catalyst only

(4) Concentrations of reactants and products and also on temperature

89. Which of the following is a Lewis acid?

(1) NH_3 (2) BF_3

(3) H_2O (4) PH_3

90. If enthalpy of combustion of carbon is – x kJ mol^{-1} then enthalpy of formation of CO_2 (in kJ mol^{-1}) is

(1) – 44 x (2) $\frac{-x}{44}$

(3) – x (4) x

BOTANY

91. For the cells of a particular tissue, we **cannot** say that these cells
    (1) Are structurally similar
    (2) Have different origin
    (3) Are structurally dissimilar
    (4) Perform a common function

92. Plant tissues are broadly classified into two main groups which are
    (1) Meristematic tissues and permanent tissues
    (2) Simple tissues and complex tissues
    (3) Ground tissues and vascular tissues
    (4) Primary tissues and secondary tissues

93. Axillary buds in plants are constituted by
    (1) Intercalary meristem
    (2) Lateral meristem
    (3) Secondary meristem
    (4) Shoot apical meristem

94. All of the following are the functions of apical meristems, **except**
    (1) Formation of secondary tissues
    (2) Elongation of plants along their axis
    (3) Primary growth of plants
    (4) Increase in the length of roots

95. Which of the following in **not** a cylindrical meristem?
    (1) Fascicular cambium
    (2) Interfascicular cambium
    (3) Apical meristem
    (4) Cork cambium

96. Apical meristems do **not** form
    (1) Dermal tissues
    (2) Ground tissues
    (3) Vascular tissues
    (4) Cork tissues

97. Major component within a plant organ is formed of
    (1) Sclerencyma (2) Parenchyma
    (3) Anucleate cells (4) Collenchyma

98. The plant cells tissues that can perform photosynthesis may be
    (1) Collenchyma and sclerenchyma
    (2) Parenchyma and collenchyma
    (3) Parenchyma and companion cells
    (4) Elongated and lignified

99. Which of the following functions is generally performed by collenchyma?
    (1) Secretion of waxy material
    (2) Storage of food
    (3) Mechanical support to the petiole of leaf
    (4) Absorption of water and minerals

100. The xylem element that is generally absent in gymnosperms is
    (1) Vessel
    (2) Xylem fibre
    (3) Xylem parenchyma
    (4) Tracheid

101. The arrangement of primary xylem is called endarch when
    (1) Protoxylem lies inside the stele
    (2) Metaxylem lies inside the stele
    (3) Both protoxylem and metaxylem form an arch like shape inside the stele
    (4) Protoxylem lies towards the centre and metaxylem towards the periphery

102. The elements of xylem that help in the radial conduction of water and can store tannins are
    (1) Xylem fibres
    (2) Tracheids
    (3) Xylem parenchyma cells
    (4) Vessels

103. Select the **correct** statement w.r.t. metaxylem.
    (1) These are formed during secondary growth
    (2) They have vessels with narrow diameter
    (3) These are late formed primary xylem
    (4) These are formed by interfascicular vascular cambium

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104. Tracheids are different from vessels as the former
(1) Lack protoplast at maturity
(2) Are elongated cells with tapering ends
(3) Have oblique perforation plates
(4) Have peripheral protoplasm at maturity

105. Which amongst the following elements of phloem remain living even at the maturity?
(a) Companion cells
(b) Sieve tube elements
(c) Bast fibres
(d) Phloem parenchyma cells
The **correct** one(s) is/are
(1) (a), (b) and (d)
(2) (a), (c) and (d)
(3) (d) only
(4) (a) and (d) only

106. The components of phloem that help in maintaining the pressure gradient in the sieve tubes are
(1) Specialised parenchymatous cells
(2) Thick walled cells
(3) Devoid of nucleus
(4) The storage site for resins, latex and mucilage

107. Which living component of phloem is/are **not** found in most of the monocotyledons?
(1) Phloem fibres
(2) Companion cells
(3) Phloem parenchyma
(4) Sieve tube elements

108. Select he **incorrect** match from the following.

(1)	Epidermal cells	–	Parenchymatous
(2)	Cuticle	–	Absent in roots
(3)	Subsidiary cells	–	Possess chloroplasts
(4)	Guard cells	–	Dumb-bell shaped in grasses

109. Identify the statement which is **not** true for trichomes in plants.
(1) They may be secretory in function
(2) They help in absorption of minerals from water and soil
(3) They help in preventing water loss due to transpiration
(4) They are usually multicellular but may be unicellular

110. The ground tissue system in angiosperms does **not** include
(1) Cortex
(2) Pericycle
(3) Pith
(4) Vascular bundle

111. The innermost layer of cortex in dicot roots is
(1) Bundle sheath
(2) Endodermis
(3) Pericycle
(4) Hypodermis

112. Presence of conjoint and open vascular bundle is the feature of
(1) Monocot stem
(2) Dicot stem
(3) Dicot root
(4) Monocot root

113. In plants, many of the cells of epiblema form
(1) Root cap
(2) Secondary roots
(3) Root hairs
(4) Root nodule

114. Which of the following statements is **not** true regarding the structure in roots that has casparian strips?
(1) It is made up of parenchymatous cells
(2) It surrounds the stele of the root
(3) It has may intercellular spaces through which water enters the vascular bundles
(4) It comprises a single layer of barrel-shaped cells

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115. During secondary growth in dicot roots, initiation of lateral roots and vascular cambium take place from
(1) Pericycle
(2) Endodermis
(3) Hypodermis
(4) Medullary rays

116. On which of the following basis we can differentiate dicot root from monocot root?
(1) Presence or absence of endodermis
(2) Number of xylem bundles
(3) Presence or absence of pith
(4) Orientation of protoxylem and metaxylem

117. Read the following features :
(a) Water containing cavities in the vascular bundle.
(b) Sclerenchymatous hypodermis.
(c) Endodermis referred to as starch sheath.
(d) Semi-lunar patches of sclerenchyma above the phloem.

Which amongst the above is/are **true** for monocot stem?
(1) (a) only (2) (a), (b) and (c)
(3) (c) and (d) (4) (a) and (b) only

118. Match the following columns and select the **correct** option.

	Column-I		**Column-II**
a.	Bulliform cells	(i)	Formed by secondary meristem
b.	Interfascicular cambium	(ii)	Found in grass leaves
c.	Secondary medullary rays	(iii)	Formed from the cells of medullary rays
d.	Heartwood	(iv)	Comprises dead elements with lignified walls

(1) a(iv), b(iii), c(i), d(ii) (2) a(ii), b(i), c(iv), d(iii)
(3) a(ii), b(iii), c(i), d(iv) (4) a(ii), b(iii), c(iv), d(i)

119. Cell theory does **not** state that
(1) All living organisms are composed of cells
(2) A newly formed cell differentiate and become specified for a specific function
(3) All cells arise from pre-existing cells
(4) Activities of an organism are outcome of activities of its constituent cells

120. The cell organelle that is found in all eukaryotic and prokaryotic cells
(1) Is only found in cytosol
(2) Is bounded by single membrane
(3) It also found in other cell organelles
(4) Does not participate in synthesis of polypeptides

121. Identify the statement as true (T) and false (F) w.r.t. the smallest cell of the living world and choose the option accordingly.
(A) It is devoid of cell wall.
(B) It can perform photosynthesis.
(C) It is prokaryotic.

	(A)	**(B)**	**(C)**
(1)	F	T	T
(2)	T	T	F
(3)	T	F	T
(4)	F	F	T

122. A protoplast does **not** include
(1) Cell wall
(2) Cell organelles
(3) Nucleus
(4) Plasma membrane

123. The below given features belong to which of the following organisms?
- Genetic material is in direct contact with cytoplasm.
- Respiratory enzymes are associated with plasma membrane.
- Ribosome are of 70S type.

(1) Fungi
(2) Prokaryotes
(3) Unicellular eukaryotes
(4) Protozoans

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124. Bacteria can be classified into two groups on the basis of the difference in the cell envelope. These are
(1) Cell walled and without cell wall
(2) Cellulosic and non cellulosic
(3) Single membraned and without membrane
(4) Gram negative and Gram positive

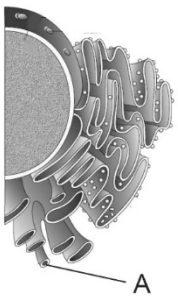
125. Bacterial flagellum is composed of
(1) Basal body and microtubules
(2) Actin and myosin proteins
(3) Filament, hook and basal body
(4) Basal body, axoneme and radial spokes

126. In bacteria, a polysome has
(1) Several ribosomes on a single mRNA
(2) Several mRNAs on a single ribosome
(3) Several ribosomes attached with each other by protein filament
(4) Several ribosomes attached to the plasma membrane

127. Select the **incorrect** statement w.r.t. plasma membrane.
(1) In the bilayer, major lipids are phospholipids
(2) All proteins associated with the membrane are buried in it
(3) Lipids are arranged within the membrane with the polar head towards the outer side
(4) Non-polar tail of lipid is protected from the aqueous environment

128. Mostly, the water molecules move across the plasma membrane
(1) From its higher concentration to the lower concentration
(2) By the expense of energy
(3) By active transport
(4) Against the concentration gradient

129. Select the option in which the cell organelle, represented as 'A' in the given figure, **correctly** matched with its function.



(1)	Rough endoplasmic reticulum	–	Synthesis of glycoproteins
(2)	Golgi apparatus	–	Formation of glycolipids
(3)	Nuclear pore	–	Controls the passage of materials inside and outside of nucleus
(4)	Smooth endoplasmic reticulum	–	Synthesis of lipid

130. In a cell, Golgi apparatus remains in close association with which of the following organelles?
(1) Mitochondria
(2) Vacuoles
(3) Ribosomes
(4) Endoplasmic reticulum

131. The membrane bound structure in the cell that contains enzymes for digesting nucleic acids is
(1) Peroxisome
(2) Contractile vacuole
(3) Nucleus
(4) Lysosome

132. The cristae of mitochondria is functionally similar to
(1) Plasma membrane of *Amoeba*
(2) Stroma of chromoplast
(3) Mesosome of bacteria
(4) Inner membrane of chloroplast

133. Enzymes for the synthesis of both carbohydrates and proteins are present in the
(1) Matrix of mitochondria
(2) Stroma of chloroplast
(3) Lumen of Golgi apparatus
(4) Cytosol of eukaryotes only

134. In which of the following chromosomes, their arms **cannot** be considered as p arm and q arm?

(a) Metacentric chromosome

(b) Sub-metacentric chromosome

(c) Acrocentric chromosome

(d) Telocentric chromosome

The **correct** one(s) is/are

(1) (a) and (b)

(2) (d) only

(3) (a) and (d)

(4) (c) and (d)

135. The RNAs that compose 80S ribosomes is actively synthesised in

(1) Cytosol

(2) Nucleolus

(3) Endoplasmic reticulum

(4) Perinuclear space

ZOOLOGY

136. Gastric juice contains all of the enzymes given below **except**

(1) Pepsin

(2) Lipase

(3) Rennin

(4) Trypsin

137. Match the items in Column-I with those in Column-II and choose the option with all **correct** match.

	Column-I		**Column-II**
a.	Cholesterol and phospholipids	(i)	Parietal cells
b.	Salivary amylase	(ii)	Succus entericus
c.	Dipeptidases	(iii)	Bile juice
d.	HCl secretion	(iv)	Parotid glands

(1) a-(i), b-(ii), c-(iii), d-(iv)

(2) a-(iv), b-(iii), c(ii), d(i)

(3) a-(iii), b-(iv), c-(ii), d(i)

(4) a-(ii), b-(iii), c-(iv), d-(i)

138. Choose the **odd** one w.r.t. accessory digestive glands in humans

(1) Gastric glands

(2) Salivary glands

(3) Liver

(4) Pancreas

139. Which of the following substance is **not** absorbed in stomach of humans?

(1) Simple sugar

(2) Water

(3) Alcohol

(4) Insulin

140. Select the **incorrect** statement w.r.t. digestion.

(1) The major components of our food digested in alimentary canal are carbohydrates, lipids, proteins and vitamins

(2) Digestion converts complex food substances into simple and absorbable forms

(3) Digestion is carried out by mechanical and biochemical processes

(4) Liver is a accessory digestive gland but does not secretes any digestive enzyme

141. In humans, the types of dentition present are

(1) Thecodont, Pleurodont, Homodont

(2) Homodont, Thecodont, Diphyodont

(3) Thecodont, Diphyodont, Heterodont

(4) Heterodont, Monophyodont, Pleurodont

142. Choose the option which fills the blank to complete the analogy.

Gastro-oesophageal sphincter : present between oesophagus and stomach : : _____ : present between stomach and small intestine.

(1) Upper oesophageal sphincter

(2) Cardiac sphincter

(3) Ileocecal valve

(4) Pyloric sphincter

143. In our alimentary canal symbiotic micro-organisms are present in a small blind sac like structure called
(1) Duodenum (2) Caecum
(3) Stomach (4) Rectum

144. Which of the following layer is exclusively present in the wall of stomach?
(1) Serosa
(2) Longitudinal muscle layer
(3) Oblique muscle layer
(4) Mucosa

145. Read following statements w.r.t. histology of alimentary canal and choose the option containing only **correct** statements
a. Serosa consists of mesothelium with underlying connective tissues.
b. All the four types of basic tissues are present in submucosa.
c. Mucosa forms irregular folds in stomach called villi.
d. Villi are supplied with a network of blood capillaries called lacteals.
(1) a and b (2) b and c
(3) c and d (4) a and d

146. Select the option that reflects the pO_2 and pCO_2 in alveolar air in comparison to atmospheric air?
(1) pO_2 lesser, pCO_2 higher
(2) pO_2 higher, pCO_2 higher
(3) pO_2 higher, pCO_2 lesser
(4) pO_2 lesser, pCO_2 lesser

147. The first step in inhalation is
(1) Transport of gases from lungs to blood
(2) Transport of gases from blood to tissues
(3) Exchange of O_2 from the atmosphere with CO_2 produced by the cells
(4) Catabolism of food substances to release energy within cells

148. Branchial respiration is present in
(1) *Periplaneta* (2) *Bungarus*
(3) *Pinctada* (4) *Aptenodytes*

149. Match the items given in Column-I with those in Column-II w.r.t. animals and their respiratory organs and choose the option with all **correct** match.

	Column-I		**Column-II**
a.	*Labeo*	(i)	Lungs
b.	*Pheretima*	(ii)	Tracheal tubes
c.	*Periplaneta*	(iii)	Moist cuticle
d.	*Pteropus*	(iv)	Gills

(1) a(i), b(ii), c(iii), d(iv) (2) a(iv), b(iii), c(ii), d(i)
(3) a(iii), b(iv), c(i), d(ii) (4) a(ii), b(i), c(iii), d(iv)

150. Trachea divides into right and left primary bronchi at the level of
(1) 7th thoracic vertebra (2) 3rd thoracic vertebra
(3) 5th thoracic vertebra (4) 7th cervical vertebra

151. All of the following structures present in our respiratory system are supported by incomplete cartilaginous rings **except**
(1) Trachea
(2) Secondary bronchi
(3) Respiratory bronchioles
(4) Initial bronchioles

152. Conducting part of respiratory system starts from
(1) External nostrils (2) Pharynx
(3) Trachea (4) Glottis

153. Which of the following structure separates thoracic cavity from abdominal cavity?
(1) Sternum (2) Thoracic vertebrae
(3) Diaphragm (4) Ribs

154. During normal inspiration, intra-pulmonary pressure decreases to generate a pressure gradient between alveoli and atmosphere. It occurs by
(1) Contraction of diaphragm and internal intercostal muscles
(2) Contraction of external and internal intercostal muscles
(3) Contraction of only diaphragm and relaxation of external intercostal muscles
(4) Contraction of diaphragm and external intercostal muscles and relaxation of internal intercostal muscles

155. Which is the common site for opening of larynx, oesophagus, eustachian tube, buccal cavity and nasal cavity?
(1) Wind pipe (2) Gullet
(3) Pharynx (4) Glottis

156. Choose the disease caused due to protein deficiency and is usually not accompanied by calorie deficiency, in a child of more than one year in age.
(1) Kwashiorkar (2) Marasmus
(3) Beri-Beri (4) Rickets

157. Read following statement A and B carefully and choose the **correct** answer.
(A) Secretion of Goblet cells protect the inner lining of gastro-intestinal tract from various enzymes
(B) HCl secreted from oxyntic cells indirectly helps in erythropoiesis
(1) Both statements A and B are correct
(2) Both statements A and B are incorrect
(3) Only statement A is correct
(4) Only statement B is correct

158. Match the following cells present in Column-A with their respective location in organs given in Column-B.

	Column-A		**Column-B**
a.	Peptic cells	(i)	Liver
b.	Paneth cells	(ii)	Pancreas
c.	Kupffer cells	(iii)	Small intestine
d.	Acinar cells	(iv)	Stomach

	a	b	c	d
(1)	i	ii	iii	iv
(2)	iii	ii	iv	i
(3)	iv	i	iii	ii
(4)	iv	iii	i	ii

159. Which of the following substances are actively absorbed in small intestine through brush border cells?
(1) Fructose and fatty acids
(2) Na^+, Cl^-, HCO_3^- and vitamins
(3) Fatty acids, glycerol and ribose sugar
(4) Na^+, glucose and glycine

160. Select **mismatch** w.r.t. accessory digestive gland and their position in human body.
(1) Parotid gland – Cheek region below ear lobule
(2) Sub-maxillary gland – At lower jaw
(3) Liver – In abdominal cavity just below diaphragm
(4) Pancreas – Above c-shaped curvature of stomach

161. Which of the following duct combines with main pancreatic duct to form common hepato-pancreatic duct?
(1) Common bile duct
(2) Cystic duct
(3) Right hepatic duct
(4) Common hepatic duct

162. How many statements given below are **incorrect**?
(a) Three pairs of salivary glands are present in buccal cavity to secrete saliva
(b) Taste buds are small projections present on upper surface of tongue which bear sensory papillae
(c) Sphincter of Oddi guards the opening of common bile duct
(d) The chemical process of digestion is initiated into oral cavity
(1) 2 (2) 4
(3) 1 (4) 3

163. Buccal cavity is responsible for all of the following functions **except**
(1) Mastication of food
(2) Facilitate swallowing
(3) Chemical digestion of food
(4) Absorption of maltose produced from starch

164. How many enzymes given in following box are secreted in their inactive form in humans?

Pepsin, Trypsin, Carboxypeptidase, Aminopeptidase, Nucleases, Nucleosidases

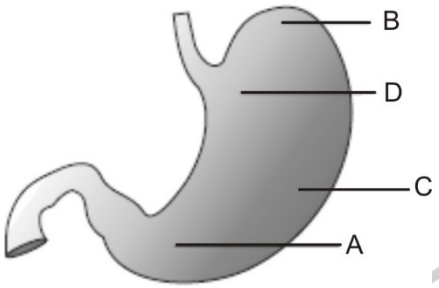
(1) 4

(2) 3

(3) 1

(4) 2

165. Anatomical regions of human stomach are shown in a given figure. Choose the option which **correctly** represents A, B, C and D.



	A	B	C	D
(1)	Fundus	Cardiac	Body	Pyloric
(2)	Pyloric	Fundus	Body	Cardiac
(3)	Body	Cardiac	Pyloric	Fundus
(4)	Cardiac	Fundus	Body	Pyloric

166. Which of the following components of our food are required in small quantities?

(1) Carbohydrates and proteins

(2) Proteins and minerals

(3) Proteins and lipids

(4) Minerals and vitamins

167. Milk sugar is digested in

(1) Stomach with the help of enzyme rennin

(2) Stomach with the help of enzyme pepsin

(3) Duodenum with the help of enzyme chymotrypsin

(4) Small intestine with the help of enzyme lactase

168. Select the **incorrect** match.

(1)	Jaundice	Defect in liver which leads to yellow colour of skin
(2)	Vomiting	Increased liquidity of the faecal discharge
(3)	Constipation	The faeces are retained within the colon due to irregular bowel movements
(4)	Indigestion	Food is not properly digested leading to a feeling of fullness

169. Site for exchange of gases in respiratory system of human include

(1) Pharynx, trachea and bronchi

(2) Alveoli and their ducts

(3) All bronchi and terminal bronchioles

(4) All bronchioles and bronchi

170. Choose the **incorrect** statement w.r.t. larynx.

(1) It is a bony box situated in the region of neck

(2) Pharynx opens into the trachea through the larynx

(3) During swallowing of food, glottis is covered by epiglottis

(4) This is a sound producing organ in both males and females

171. Which of the following is **correct** sequence of events for inspiration?

A. Contraction of diaphragm and external intercostal muscles increase the volume of thoracic chamber anterioposterior and dorsoventral axis respectively.

B. Air rushes into lungs under negative pressure breathing.

C. Intrapulmonary pressure decreases.

(1) A, B, C (2) A, C, B

(3) B, A, C (4) B, C, A

172. Ratio of tidal volume to inspiratory reserve volume is approximately equal to ratio of

(1) TV to VC (2) TLC to FRC

(3) RV to TLC (4) EC to VC

173. A healthy man can inspire or expire approximately _____ ml air per minute. Choose the option which fills the blank **correctly**.
(1) 500 - 600 (2) 600 - 800
(3) 800 - 1000 (4) 6000 - 8000

174. Select the blood vessel among the following which carries least CO_2 in our body?
(1) Vena cava
(2) Hepatic vein
(3) Pulmonary vein
(4) Pulmonary artery

175. Maximum volume of air that a person can expire after maximum inspiration is represented as
(1) Tidal volume
(2) Vital capacity
(3) Inspiratory capacity
(4) Expiratory reserve volume

176. At the tissue level, pCO_2 is high due to metabolism and it diffuses into RBCs and increases H^+ concentration as represented by following reversible reaction ;

$$CO_2 + H_2O \underset{\text{anhydrase}}{\overset{\text{carbonic}}{\rightleftharpoons}} H_2CO_3 \underset{\text{anhydrase}}{\overset{\text{carbonic}}{\rightleftharpoons}} HCO_3^- + H^+$$

What will happen at alveolar surface where the above reaction is reversed?
(1) Blood at alveolar level contains more chloride ions within RBCs
(2) pH of oxygenated blood will be more due to more hydrogen ions
(3) pH of oxygenated blood will increase due to removal of CO_2 in expired air
(4) Oxygen dissociation curve will shift towards right side

177. Which of the following condition does **not** favour the formation of oxyhemoglobin at alveolar site?
(1) High partial pressure of O_2
(2) High pH of blood
(3) Low partial pressure of CO_2
(4) Higher temperature

178. Which of the following factors favour decrease in breathing rate?
(1) Increased blood oxygen
(2) Increase blood CO_2
(3) Increased metabolism
(4) Increase in acidity of body fluid

179. The switch off point of breathing which can moderate the function of respiratory rhythm centre in medulla is situated in
(1) Pons (2) Cerebrum
(3) Cerebellum (4) Midbrain

180. Select the **mismatch.**

(1)	Asthma	Allergic disease characterised by inflammation of respiratory tract which leads to wheezing
(2)	Emphysema	Caused mainly due to chronic cigarette smoking
(3)	Asbestosis	Leads to fibrosis of lungs
(4)	Pleurisy	Inflammation of alveoli due to water pollution

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